

Hydro and Wind Power: Complements or Substitutes?

Paper Discussion

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- **Dynamic stochastic model** of a Finnish system operator, **schedules and dispatches generation** hour-by-hour to maximize expected social surplus.
- **Hourly two-stage decision process**
 - First: Forecasting of load, wind, CHP and hydro inflow → schedule fossil fuel generators.
 - Second: After realization of these key variables → dispatch hydro and fossil capacity.
 - Setup with three generation sources: Thermal plants (merit-order), **wind** (exogenous capacity and weather-driven capacity factor), **hydro** (flexible reservoir stock and RoR).
- **Weather variables** and **load/generation** estimated in SUR system, hourly 2022 Finish data.
- Dynamic model, adjustments in two "dimension"
 - Scaling up wind capacity by +50% and +100% (of 5'000 MW)
 - +/-10% and +/-30% rescale of hydro
 - Asses social benefits (operating cost, fossil reserves, price, outage probability, CO2 emissions)

Main Comments

- Rescaling hydro **including run-of-river** by adjusting must-run nuclear → substitute a flexible for an inflexible resource.
 - *Higher RoR generation exacerbates wind curtailment in spring/summer.*
- **Exclusion of Nord Pool** prohibits export of excess wind to neighbors.
- Using **2022 as a single weather year**
 - *Other years as robustness check.*
 - *Estimating 8 equation on 8760 hours, might not include 100 years weather variability.*
- Ext: Assumed **no price / curtailment response** of consumers
 - Contrast to Gowrisankaran et al. 2016; EV 48% of new Finish cars in 2026.
 - *One aggregated battery / EV component (similar to aggregated reservoir)*
 - *Might allow for higher wind penetration.*
- Ext: **Data based (but conservative) emission price** of 80 EUR/tCO₂.
 - *Sensitivity analysis for larger range in social cost of carbon .*
 - *Higher wind share more beneficial, push substitute threshold further upwards.*

- More runs to show smaller change increments (for hydro and wind)
- Converting outcome values (e.g. Table 5, p34) to a line graph (serves complement / substitute story + show "optimal" state).
- Single hydro reservoir but explicit mention of individual reservoir specific multiplier (p18).
- The paper refers to wind capacity increases of 50% and 100% without clearly stating the baseline capacity (5'000 MW) at first mention, making the interpretation of the counterfactuals harder at a first glance.